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## **Experiment 04: Informed Search**

**Learning Objective:** Student should be able to solve a given problem using Informed search technique.

**Tools:** Python under Windows or Linux environment

**Theory:** Study and implement Best First Search or A\* search under Informed Search techniques.

**ALGORITHM:** (for selected search technique)



**ADVANTAGES:**

1. ....
2. ....

**DISADVANTAGES:**

1. ....
2. ....

**PROPERTIES:**

**Complete:** .....

**Time:**  
.....

**Space:** .....

**Optimal:** .....

**APPLICATIONS:**

1. ....
2. ....

**Design:** .....

**Implementation:**

**Result and discussion:** .....

**Learning Outcomes:** Students should have the ability to

LO1: identify a problem which can be solved using uninformed search methods.

LO2: implement uninformed search methods.

LO3: describe properties of uninformed search algorithm.

LO4: identify advantage and disadvantage of the algorithm.

**Course Outcomes:** Upon completion of the course students will be able to evaluate various problem solving methods for an agent to find a sequence of actions to reach the goal state.

**Conclusion:** .....

**Viva Questions:**

1. Describe the structure of a search space in which iterative deepening search performs much worse than DFS.
2. Name which algorithm overcomes drawback of DFS and BFS?
3. Compare and contrast uninformed search strategies with respect to solving 8 puzzle problem.

For Faculty Use

| Correction Parameters | Formative Assessment [40%] | Timely completion of Practical [ 40%] | Attendance / Learning Attitude [20%] |  |
|-----------------------|----------------------------|---------------------------------------|--------------------------------------|--|
| Marks Obtained        |                            |                                       |                                      |  |